NTNU

Kunnskap for en bedre verden

TDT4259 - Applied Data Science

Day-ahead Electricity Spot Price Forecasting

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Introduction

This report is the final product of the work done during the group assignment for the course TDT4259 - Applied Data Science at the Norwegian University of Science and Technology (NTNU).

In the following paper there have been outlined all the main tasks to undergo to complete the project, from a brief introduction, to a description of the methods to be used and the evaluation and interpretation of the produced results, while also covering a possible deployment plan of the final product on the market and the monitoring process of the implemented pipeline after deployment.

1 Context and Problem Definition

This section outlines the context and defines the problem, concluding with an overview of the team and their roles.

1.1 SINTEF & Nord Pool

SINTEF (*Stiftelsen for industriell og teknisk forskning*, which translates to "The Foundation for Industrial and Technical Research") is one of Europe's largest non-profit independent research institutions. Originally created in 1950 by the Norwegian University of Science and Technology, intended as link between the university and the industry, the establishment quickly expanded and now counts over 2000 employees and 3 billion NOK [1].

Based in Trondheim, SINTEF has offices all over Norway, with laboratories ranging from microelectronics and nanoscale technologies to oil and gas transport and the maritime industry [2]. One of SINTEF's recent most important enterprises is focusing on advancing the integration of diverse energy carriers for sustainability. To do that, they have been focusing on developing a model to forecast electricity spot prices in the Nord Pool market (Figure 1) [3], Europe's first power exchange using historical spot price and volume data, and that's what's going to be the aim of this project.

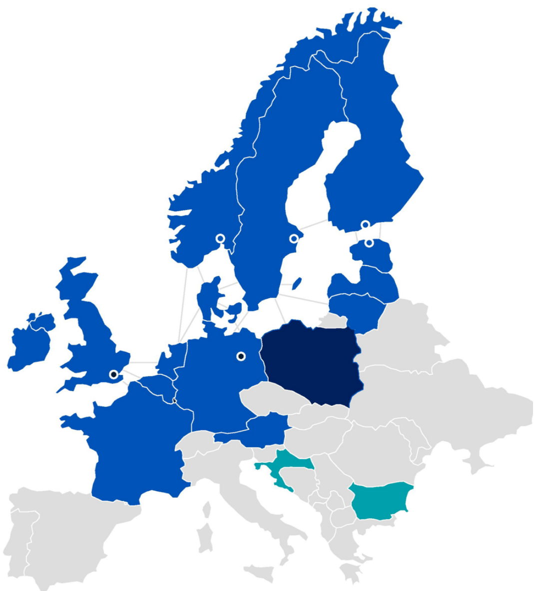

Figure 1: Nord Pool market

1.2 Problem Definition

The main goal of this project is the development of a machine learning model which is able to effectively forecast electricity spot prices in the Nord Pool market. This market, even if is deeply integrated and very efficient, is also one of the most volatile and unpredictable. This is due to the high dependence of Scandinavian countries on renewable resources. Hydroelectric and wind energy ensure that Northern Europe relies little on fossil fuels, making the energy production very difficult to predict, as it strongly depends on weather conditions. This market, as any other financial market, owes its complexity also to the critical roles played by geopolitical events, economic trends, and regulatory changes.

Nord Pool operates as a day-ahead and intra-day market where electricity is traded based on forecasted supply and demand. It is important to keep in mind that the electricity spot price refers to the price at which electricity is bought and sold for immediate delivery in a specific time frame. In financial markets, spot price contrasts with forward/future price, which is a contracted price for a transaction to be completed at an agreed-upon future date.

Being able to effectively predict electricity price may enable better decision-making for stakeholders actively involved in the energy sector, as producers and consumers. Understanding future price movements allows improvements in bidding and hedging strategies, as well as exploring market efficiency. In particular, this work is focused on an active player in the Nord Pool market, such as a stakeholder who buys and sells energy.

1.3 Importance of the Analysis

As already stated above, an effective forecast of electricity spot prices is very valuable for the decision-makers involved in the energy trading system. There are several reasons of this, and, for the type of stakeholders defined for this project, in the following sections there have been outlined Market Efficiency & Strategic Planning and Risk Management, as these are typically key concerns for market participants.

Market Efficiency & Strategic Planning

In financial terms, market efficiency refers to the degree to which market prices reflect all available, relevant information. If markets are efficient, then all information is already incorporated into prices, and so there is no way to "beat" the market because there are no undervalued or overvalued securities available. As the quality and amount of information increases, the market becomes more efficient reducing opportunities for arbitrage and above market returns [4].

Participants in electricity market can enhance their bidding strategies if they are supported by reliable price forecasts. In particular, the understanding of potential price movements can help them make informed decisions about when to buy or sell electricity, leading to competitive bidding and reduced volatility in market prices. Moreover, regulatory agencies can use price predictions to monitor market behavior and implement policies that ensure fair competition and prevent market manipulation. Thanks to this insights into expected price trends, regulators can better assess the impacts of market interventions and maintain a stable operating environment. An efficient and reliable market can provide its actors with different advantages:

1. All significant information is available and already integrated in market prices. In this way, participants are able to make informed decisions, minimizing informative asymmetry;
2. Efficient markets are characterized by high liquidity level. This reduces bid-ask spreads and the transaction costs;
3. Prices reflect the real value of assets, thanks to the speed of information dissemination. Consequently, it is less likely that misalignment will occur between the price and the intrinsic value of the asset.

Risk Management

The second challenge is related to the fact that the companies involved in financial markets always need to develop strategies in order to manage the risks in trading. They can implement more effective hedging strategies when they can accurately predict spot prices. This capability reduces the financial risks associated with price volatility, providing a buffer against unforeseen market fluctuations.

For example, an electricity producer anticipates that the spot price of electricity will decrease in the future due to seasonal changes. To protect against this potential price drop, the producer can enter into a futures contract to sell electricity at a predetermined price. This is just the simplest example of a possible hedging strategy, considering that many other complex derivative contracts can be used in this market.

1.4 Team, Roles, and Responsibilities

The team for this project consists of a diverse group of students from different universities from Italy, currently participating in the Erasmus program at the Norwegian University of Science and Technology (NTNU). The following Table 1 provides an overview of each team member's expertise and the roles they will play in the project.

Member	**Academic background**	**Responsibilities**
Amadori Luca	MSc student in Quantitative Finance - Mathematical Engineering at Politecnico di Milano	Understanding the business context. Price analysis to deepen insights into the dynamics of this specific financial market. Support for model implementation and analysis of results.
Coppola Rodolfo Emanuele	MSc student in Quantitative Finance - Mathematical Engineering at Politecnico di Milano	Business understanding and possible trading/hedging implications. Analysis of price and returns distributions, linked to the comprehension of this specific financial market. Support to the model implementation and results analysis.
Deidier Simone	MSc student in Computer Science and Engineering at Politenico di Milano	Data analysis and visualization, dashboard's GUI development, analysis and definition of KPIs, risk management and contingency plan.
Gimmillaro Gabriele	MSc student in Statistical Learning - Mathematical Engineering at Politecnico di Milano	Error metrics choice and description, deployment and recommendations plan.
Miccio Gian Marco	MSc student in Statistical Learning - Mathematical Engineering at Politecnico di Milano	Data analysis, interpretation, and visualization; deployment and recommendations plan. Color palette designer.
Peiretti Riccardo	MSc student in Data Science at Università di Trento	Data analysis and interpretation, data visualization, and technical aspects of the implementation of machine learning models.

Table 1: Team members and responsibilities

1.5 AI Tools

In the following Table 2, it is declare which AI tools are used in what capacity in this project.

AI Tool	How
Grammarly	Spell checking
ChatGPT	Writing assistance to have a better formal language and help in the data visualization phase to achieve a better resolution of the plots

Table 2: AI Tools

2 Background

This section outlines the objectives and purpose of the work. Furthermore, it provides a description of the well-structured data strategy employed, which is essential for producing valuable results for the company. Additionally, the design thinking method is used to complement the data strategy and to develop solutions that align with both business goals and stakeholder needs.

2.1 Objectives

In this section specific business and technical objectives of the projects are presented. Moreover, the background reasons and the methods are illustrated.

Why?

Effective forecasts of spot prices are very valuable for different reasons and in various fields, as they could shape better decision-making processes and improve energy demand balancing through coordination among factories, industries, and large corporations. In this way, it could also stabilize the use of renewable resources, marking a new step forward in the European ecological transition, recently updated through the Green Deal [5]. A good prediction model can provide an energy trader with numerous competitive advantages. In particular, it should lead to optimized bidding strategies and risk coverages tactics.

What?

The main goals of the work can be divided in two sections: **Technical** vs **Business** objectives.

Starting from the **technical** ones, three targets for the error metrics are set:

Metric	Target
MAE (Mean Absolute Error)	< 1.5
MAPE (Mean Absolute Percentage Error)	$< 4\%$
RMSE (Root Mean Square Deviation)	< 2.5

Table 3: Performance targets for prediction model

To deliver an optimal final product for the client, it is essential to meet not only the target metrics such as MAE, MAPE, and RMSE, but also additional standards like model robustness, processing speed, and scalability. These indicators ensure that the model not only maintains accurate and stable performance, but can also handle increasing workloads and provide timely responses across various operational scenarios, meeting the stakeholder's expectations and long-term needs.

Further objectives, marked under the **business** area, are pursued. In particular, the model should be able to meet the stakeholder's needs and expectations. That is, providing effective predictions which can maximize the revenues coming from the trading activity, and at the same time, minimizing the risks linked to market activity. In this sense, fast and reactive forecasts should enable the development of tailor-made hedging strategies, highly decreasing the risks encountered by the energy trader.

How?
The problem is addressed with the development of a machine learning model, possibly able to generate effective forecasts of 24h-ahead electricity spot prices. In order to increase the model's performances, different data analysis and feature engineering techniques are used. Moreover, three modeling configuration, such as Linear Regression, XGBoost, and Random Forest, are investigated [6]. The final product for the stakeholder is decided on the basis of the three error metrics: MAE, MAPE, and RMSE [7].

2.2 Data Strategy

CRISP-DM

To achieve the objectives described in the Subection 2.1, a data strategy is necessary. It aligns with the Cross-Industry Standard Process for Data Mining (CRISP-DM) [8]. It is an open standard developed in 1996 by leading companies in data analysis. CRISP-DM is the most widely used data-mining model due to its flexibility and ability to address key challenges in the industry. Its success is largely due to being neutral to industry, tools, and applications, making it adaptable across various fields [9]. The intention of the model is "to provide structured approach to planning a data mining project and to help ensure that the project is on track." [10]. CRISP-DM provides an overview of the life cycle of the work and it contains the phases, their respective tasks, and their outputs [11]. As reported in Figure 2, this approach is structured into six phases which are described in the following sections [12].

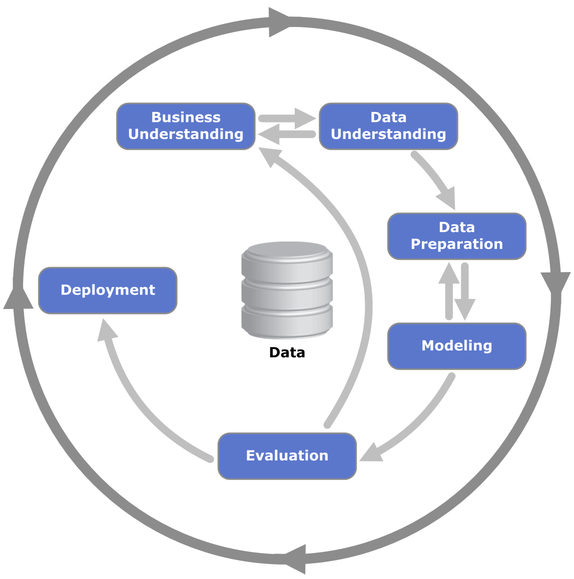

Figure 2: CRISP-DM Diagram

Business Understanding

In order to design an efficient solution for the data science project, it is essential to understand the business needs related to the project itself. In particular, there are two main goals that should be achieved in the business understanding phase [13], detailed in Subsection 2.1:

- **Determine business objectives**: understand what the client or the company want to achieve from a business point of view and define the criteria for the business success;
- **Determine data science goal**: point out the technical objectives and define the data science success criteria.

Data Understanding

This phase is crucial to kick start the project in the right direction. It involves the following:

- **Data collection**: collecting or acquiring the dataset, in this case it is used the Nord Pool dataset;
- **Data description**: getting an idea of data amount, the types of variables in which the data is stored, and necessary coding schemes to do that, process that has been explored in Subsection 3.1;
- **Data quality check**: it is important to check for missing values, inconsistencies, unintuitive data, and general errors in the code and the dataset;
- **Data exploration**: it can be summarized as everything done in Subsection 3.3, general data analysis on the singular variables, plotting the variables over time, and observing the correlation coefficients between pair of features.

Data Preparation

After understanding the available data, as described in Subsection 3.4, to prepare the data for the machine learning model. This process includes clean and pre-process the data to ensure that it is in an optimal state for model training and prediction.

- **Data section**: select features and understand the target variable;
- **Data cleaning**: correct data errors, fill in or infer missing data to make data consistent;
- **Data construction and formatting**: creating new features or modifying existing ones to better capture the underlying patterns in the data, while addressing the challenges identified during the data understanding and cleaning phases.

Modeling

The prepared data are the starting point for the next stage of the pipeline, the modeling. This phase is composed by three main tasks, that are outlined in detail in Subsection 3.5 and 4.1: s

- **Model selection**: select the most suitable algorithm to solve the problem;
- **Model building**: use techniques such as feature selection and hyperparameter optimization to enhance the predictive power of the selected model;
- **Model assessment**: evaluate the model performance on unseen data (test set), time the code and keep track of Key Performance Indicators (KPI's).

Evaluation

After the first rundown of modeling and processing, it is important to evaluate the current results available after the first work session, as reported in Section 4:

- **Results evaluation**: to understand what is the best model which meets objective of the business;
- **Process review**: a thorough review to identify overlooked factors, quality assurance issues, and missed or repeatable activities;
- **Next steps assessment**: outline possible actions with pros and cons to decide on deployment and new iterations.

Deployment

Section of the project dedicated to actually putting the final product on the market, by selling it and interacting with clients:

- **Pre-deployment**: any last tasks to be completed before launch;
- **Online testing**: making sure the APIs and websites created for user interaction work without issues;
- **Monitoring and logging**: taking into account and observing usage of the product and keeping track of any issues and anything else of note;
- **Active feedback**: establishing a feedback channel between the team and the client, to ensure;
- **On-call responsibilities**: making sure that the team is able to work live, on-demand depending on the task that needs to be addressed.

All of these points are explored in detail in Section 5.

Maintenance and Monitoring

This final phase outlines the processes designed to ensure the sustained performance, reliability, and robustness of the forecasting model for the Nord Pool market, this includes:

- **Key Performance Indicators (KPI)**: defines the metrics essential for evaluating the model's effectiveness and business impact;
- **Monitoring dashboard**: provides a visual overview of the model's performance and key metrics, specifically tailored to track electricity spot prices in the Nord Pool market;
- **Strategic risk management and contingency plan**: outlines the framework for identifying, assessing, and mitigating potential risks associated with the deployment and operation of the forecasting model.

2.3 Design Thinking

Design Thinking is a human-centered approach that emphasizes understanding users' needs, generating creative solutions, and iterating based on feedback. In the context of the Nord Pool dataset, which contains daily energy consumption, production, and cost data from the Nord Pool market, Design Thinking ensures that the project team takes the time to explore and address the specific needs of stakeholders before jumping into product development. This methodology is structured into five phases: *empathize*, *define*, *ideate*, *prototype*, and *test*.

- **Empathize**: in this initial phase, the focus is on understanding the users — energy providers, regulators, and consumers within the Nord Pool market — by gathering insights on their needs, pain points, and behaviors. For example, how do fluctuations in energy production or price volatility affect their decisions?
- **Define**: based on the insights from the empathize phase, the team defines the core problems to be addressed. For instance, how can energy consumption forecasts be made more reliable, or how can pricing models better adapt to renewable energy variability?
- **Ideate**: the aim is to generate a wide range of creative solutions to the defined problems. This could involve brainstorming new ways to visualize consumption data, or exploring different algorithms for optimizing energy production and pricing strategies;
- **Prototype**: the most promising ideas are turned into prototypes - simplified models or simulations that can be tested with users. For example, the team could develop a dashboard for energy providers to simulate different pricing models or consumption patterns;

- **Test**: the prototypes are tested with real users, and feedback is gathered to refine the solutions. In this case, energy stakeholders from the Nord Pool market might provide input on how well the prototypes meet their needs and what adjustments are necessary.

Design Thinking ensures that the project moves through these iterative stages, refining solutions based on real-world user feedback. This approach complements the CRISP-DM - reported in Section 2.2 - model by ensuring that the technical insights derived from data analysis are not only accurate but also user-centered and practical. By integrating empathy and creativity into the problem-solving process, Design Thinking allows the project to produce solutions that are both innovative and tailored to the specific needs of the Nord Pool market. Whether addressing challenges like energy price fluctuations, production variability, or policy decisions, this methodology helps ensure the end product is aligned with stakeholder goals and has meaningful real-world impact.

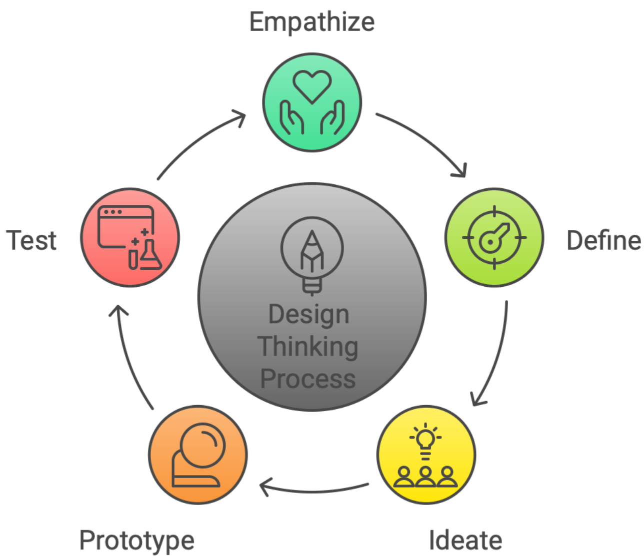

Figure 3: Phases of the Design Thinking process

3 Method and Analysis

3.1 Dataset Description

The analysis is based on the Nord Pool dataset containing the following information, collected hourly from 31-12-2015 to 13-09-2018:

1. `datetime_utc`: day and time of the observation, measured with the worldwide reference of Coordinated Universal Time (UTC);
2. `volume_demand`: it is the total request of electricity in the Nord Pool for that specific time, expressed in megawatt [MW];
3. `volume_production`: it is the total amount of electricity supplied by players involved in the Nord Pool for that specific time slot, again expressed in megawatt [MW];
4. `spot_price`: the electricity spot price refers to the price at which electricity is bought and sold for immediate delivery in a specific time frame [cents/kWh].

3.2 Methods and Tools

Throughout the project, a variety of tools have been utilized. This section frames how they are applied within the scope of the project.

Programming Language and Environment

1. **Python**: Python serves as the programming language for the entire project. The ecosystem of libraries available in Python allows to integrate a lot of functionalities useful for processing the dataset and analyzing data;

2. **Google Colab**: as an online Jupyter notebook environment, Google Colab provides an easy access to data files and collaboration with team members in real time. Moreover, it offers free access to GPUs and TPUs made it advantageous for training the models;

3. **Google Drive**: Google Drive is utilized for storing and sharing all the useful project-related files.

Data Manipulation and Processing

1. **Pandas**: Pandas is an excellent and fundamental library for data manipulation. It provides DataFrame structures that simplify to manage the data and make it easier to prepare data efficiently for modeling;

2. **NumPy**: NumPy is necessary for performing mathematical operations on large datasets during the pre-processing stage;

3. **OS**: the Operating System Library in Python is fundamental to manage file paths and directories, especially when working in a group with shared files and folders;

4. **Astral**: The astral library is a Python package used for calculating sunrise, sunset, and other solar times based on a given location. It is used to create the new feature `light_status`.

Data Visualization

1. **Matplotlib**: Matplotlib is used for creating plots and charts for understanding data distributions and trends, facilitating exploratory data analysis (EDA);

2. **Seaborn**: built on Matplotlib, Seaborn provided a higher-level interface for producing a better color visualization;

3. **Moqups** and **draw.io**: online free tool to easily develop GUI for web applications used for the application's dashboard.

Model Development

1. **Scikit-learn**: Scikit-learn is the library used for machine learning tasks, including data pre-processing and model evaluation;

2. **Linear Regression**, **XGBoost**, and **Random Forest**: for model creation, a combination of machine learning algorithms is employed to determine the best solution for this data-driven problem. The selection includes Linear Regression, XGBoost, and Random Forest, as it is discussed in the Subection 4.4.

3.3 Data Analysis

To gain a first understanding of the dataset, a preliminary data analysis has been conducted, with the aim of uncovering patterns and relationships between the variables and within the data itself. This initial analysis serves as the foundation for the subsequent advanced modeling that will be done:

- a **Correlation Matrix** was generated to identify relationships between variables, revealing potential linear dependencies between price, volume demand, and volume production;

- The **Price Distribution** was analyzed to understand the variability, range, peaks and other characteristics of energy prices, a crucial part for identifying trends and possible anomalies. A visualization was created to track changes in **price over time**, offering a perspective that takes time into consideration to highlight seasonal changes in behavior;
- The **Analysis of Price, Demand, and Production dynamics** was conducted to understand how supply and demand dynamics influence price in the dataset and hence in the market;
- Lastly, **Price, Volume Demand and Volume Production trends** over time are produced to smooth out short term high fluctuations in the data and put emphasis more on longer term trends. This approach provides a clearer visualization of general trends in the market, something that is essential for the project.

Correlation Matrix

Figure 4 shows the correlation matrix between all the variables in the dataset, a grid visualization of Pearson's correlation coefficient, which is a value that ranges from -1 to 1 and that represents the linear correlation between two sets of data. A coefficient close to 1 means that the second variable increases if the first one does, while a coefficient close to -1 means that the second variable decreases as the first one increases. Instead, a value of 0 suggests the presence of no relationship between the two variables.

At the beginning, the correlation matrix is a 4x4 sparse matrix, with correlations limited to demand volume-production volume and price-time. However, after executing feature engineering methods described in Subsection 3.4, where additional variables are obtained, new correlations are shown in Figure 15. The correlation matrix is a useful tool to identify redundant variables, as highly correlated variables carry the same information for the scope of statistical analysis.

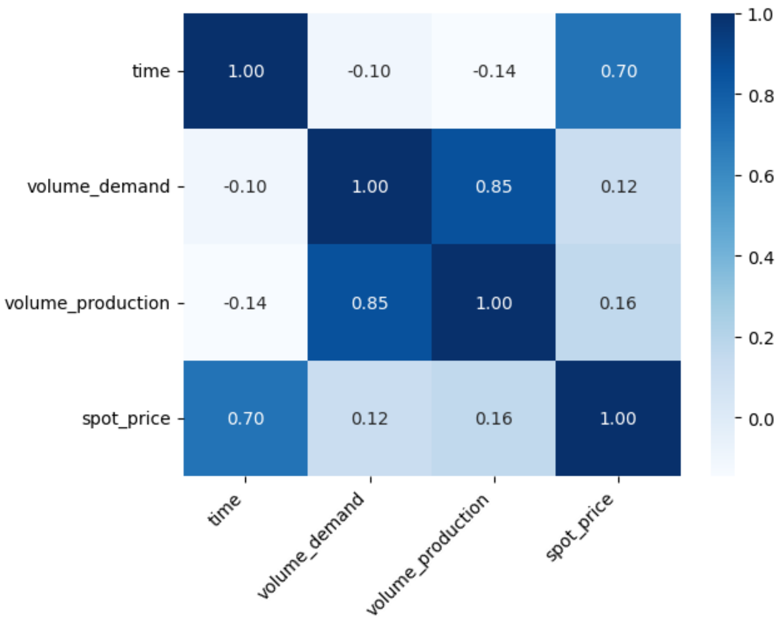

Figure 4: Correlation Matrix of the initial dataset

Price Distribution

Figure 5 above illustrates the distribution of energy spot prices in the Nord Pool market. The histogram shows a strong concentration of prices within the range of approximately 10 to 40 units, with the highest frequency observed around 20–30 units. This indicates that the majority of prices are clustered within this range. The distribution is right-skewed, with a long tail extending towards higher prices, suggesting occasional spikes in prices, though these are relatively rare compared to the main cluster around the lower price range. The smooth density curve overlay highlights the main modes and emphasizes the overall shape of the distribution, indicating that while higher prices are possible, they occur infrequently in this dataset.

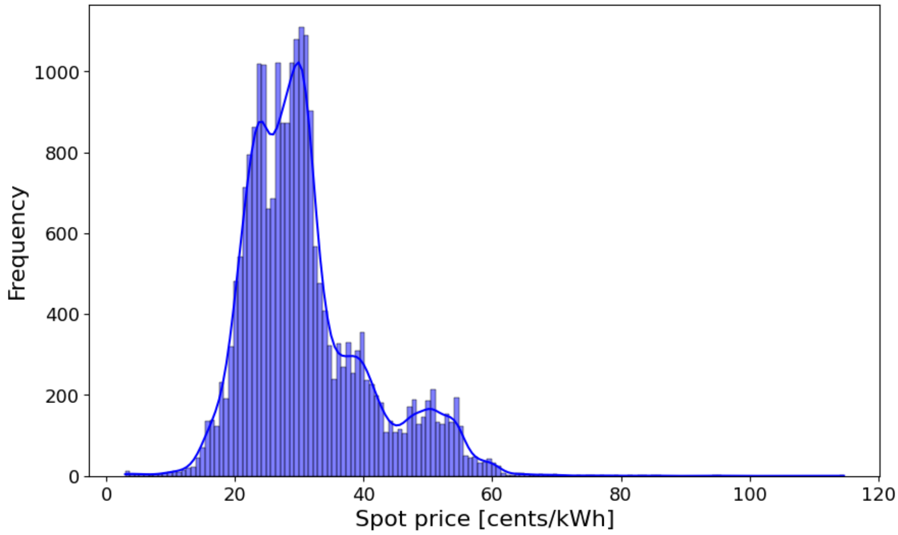

Figure 5: Distribution of price values in the dataset.

Yields Distribution

To gain deeper insights into the fluctuations of spot prices in this specific market, the frequency distribution of hourly log-returns is illustrated above. Surprisingly, the shape of this distribution lacks the fat tails typically experienced in financial markets, which usually indicate high volatility and extreme price movements. This suggests that the Nord Pool market is characterized by lower volatility, meaning price changes tend to be more stable and less prone to sudden, large swings over time. Such stability could be attributed to the regulated nature of the market, the balance of supply and demand, or specific factors inherent to energy markets, distinguishing it from more speculative financial markets.

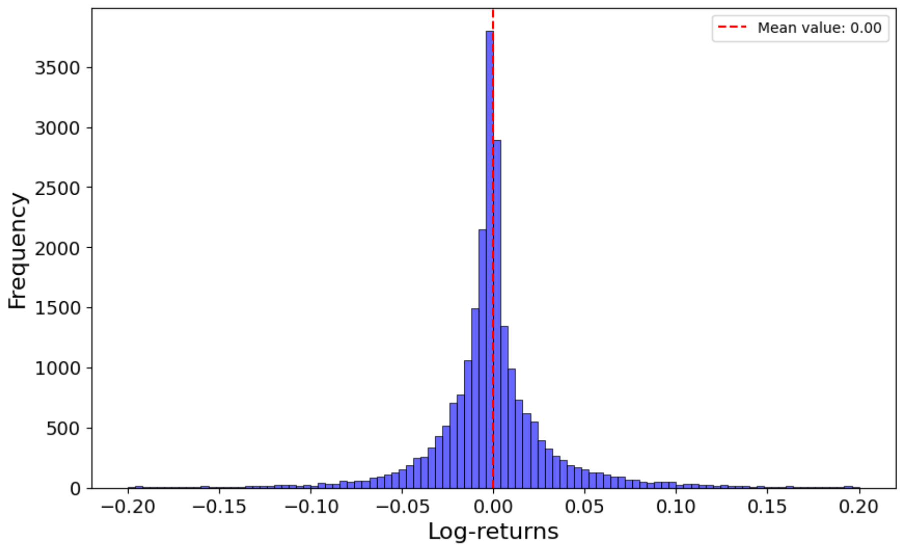

Figure 6: Hourly log-returns

Analysis of Price, Demand, and Production dynamics

This subsection explores key relationships between energy price, volume demand, and volume production within the Nord Pool market, while also taking time into account in these relationships. By examining these interactions, the aim is to understand how market dynamics affect energy pricing and supply. Three primary analysis are conducted. First, the relationship between **volume production and price** (Figure 8) is analyzed to assess how changes in production impact spot prices, together with **volume demand against price** (Figure 7), both of which giving insights on how demand fluctuations correlate with price movements.

Important note about these plots: they contain exactly one year of data from the original dataset, precisely the entirety of 2016, and the color gradient represents how far into the year the observation is taken: deep blue points are taken from January, while light green points are taken from December. It is very easy to assess the presence of time-dependent relationships between the variables, as the plot of price against one of the other two behaves very differently in a cold month compare to a warm month. This further underlines the importance of the period of the year one finds themselves in when considering the behavior of energy prices.

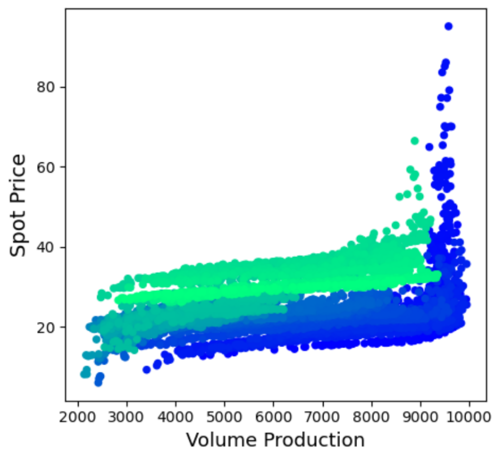

Figure 7: Volume Demand against Spot Price

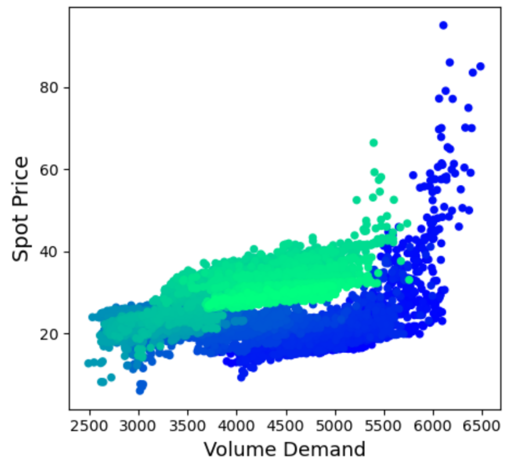

Figure 8: Volume Production against Spot Price

Secondly, by observing a time-series plot of **volume production against volume demand** (Figure 9) presented here, it is possible to highlight a linear relationship between the two variables, that is also independent of time, as there is no clear difference between the different months of the year.

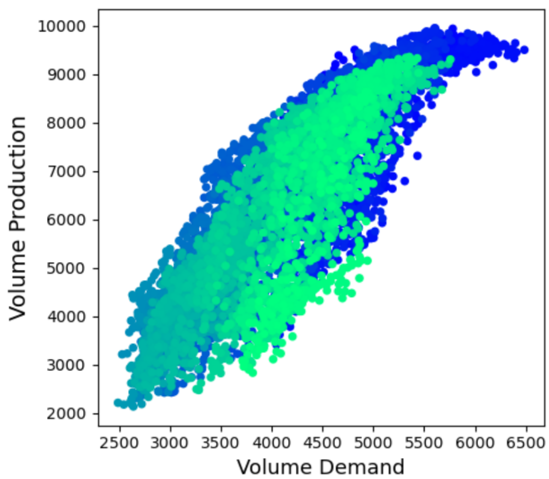

Figure 9: Volume Production against Volume Demand

Finally, the focus is shifted to a specific month's data. The following two graphs highlight the relationship between between `volume_demand` and `spot_price` during two different months, January 2016 and August 2016, while also taking into consideration the feature `light_status`, a categorical variable that represents the amount of light present at a specific point in time, further explained in 3.4. It is possible to notice a clear difference between the behavior of the variables in the two time periods, dictated by differences in the amount of day light in each of the months, temperature and climate.

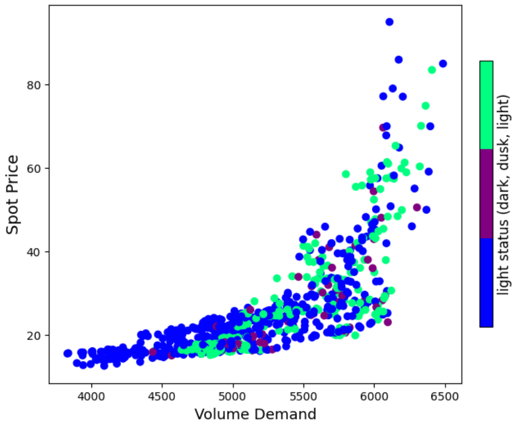

Figure 10: Volume Demand against Spot Price for the month of January 2016

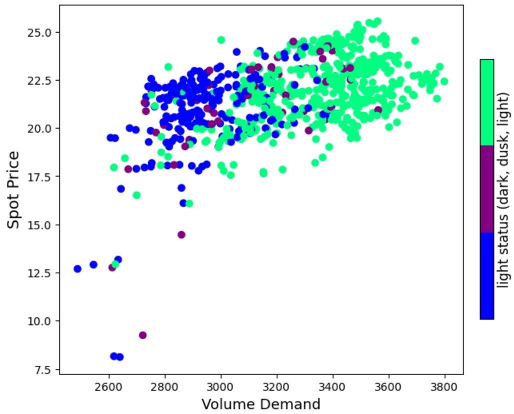

Figure 11: Volume Demand against Spot Price for the month of August 2016

These analysis are essential for understanding the factors driving price variability and the interplay between supply and demand in the Nord Pool market.

Price, Volume Demand and Volume Production trends

Figure 12 shows the **7-day moving averages** of energy **price**, **volume demand**, and **volume production** in the Nord Pool market, with each variable normalized to a range between 0 and 1. By applying a weekly moving average, short-term fluctuations are smoothed out, highlighting long-term trends and seasonal patterns in the data. Normalization enables a direct comparison of the three variables on a common scale, allowing correlations and divergences between price, demand, and production over time to be observed.

The demand and production curves (green and red, respectively) exhibit a generally similar movement, suggesting a strong correlation between the volume of energy demanded and produced. Both curves tend to rise and fall in tandem, indicating that production is responsive to demand trends, likely in efforts to maintain supply-demand balance in the market.

In contrast, the price curve (blue) displays a more spiky behavior, with sharper peaks and valleys, reflecting its higher sensitivity to rapid changes in market conditions. Unlike demand and production, price fluctuations appear more volatile and less consistent with the trends in demand and production, suggesting that price is influenced by additional factors, such as external market dynamics or supply constraints. This volatility underscores the complex nature of price formation in the energy market, where even small shifts in demand or supply can lead to pronounced changes in price.

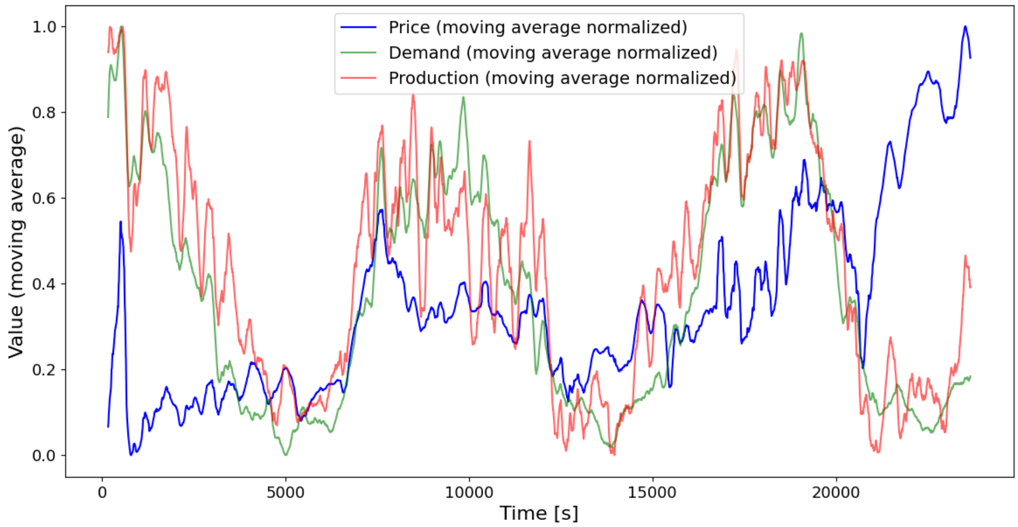

Figure 12: Price, Volume Demand, and Volume Production trends

3.4 Data Pre-processing

Data Cleaning

A brief check of the number of `NaN` values and of the number of duplicates proved that the dataset was already clean, so no data cleaning was needed. Additionally, since the dataset only includes non-negative values for prices and volumes, meaning there was no need to eliminate any rows to be considered invalid.

Feature Engineering

Feature engineering is a fundamental step of any project that involves any form of data analysis, as it helps to fully exploit the information contained in the features that might have been hidden before, and create much more effective models as a result.

1. First of all, a modification to the variable `datetime_utc` was made, by introducing `time`: it represents the time difference between each observation and the first one available in the dataset, measured in seconds [s];
2. A variable d_t was introduced with values ranging from 0 to 6 to represent the `day_of_week` of the observation. This variable is then decomposed into two new features, `day_of_week_sin` and `day_of_week_cos`, computed as the *sine* and *cosine* of $\frac{2*\pi*d_t}{6}$, in order to obtain a continuous and cyclic representation of the days of the week. This kind of feature generation takes inspiration from the **Data Analysis** performed on the data, which highlights a periodic behavior of the spot price during the week (Figure 13);
3. The new variable `spot_price_lag_24` representing the spot price at the same hour of the previous day was built. The price lagged by 24 hours serves as a straightforward predictor for the current day's price, making it a valuable inclusion to improve the model's performance;
4. In order to keep track of light role in price prediction, a function has been developed that, given a timestamp, returns 0 if it was dark, 1 if it was twilight, and 2 if there was light at that specific moment in time, using historical data measured in Oslo. The variable `light_status` may play a significant role in forecasting spot prices as it reflects the availability of natural light, which can influence a lot both energy demand and generation. During daylight hours, the demand for electricity may be lower due to increased solar power generation and reduced need for artificial lighting. Conversely, during twilight or nighttime, energy demand typically rises as people turn on lights and appliances, potentially leading to higher spot prices. Understanding the variations in `light_status` allows for better modeling of supply and demand dynamics in the energy market, leading to more accurate predictions of spot prices;

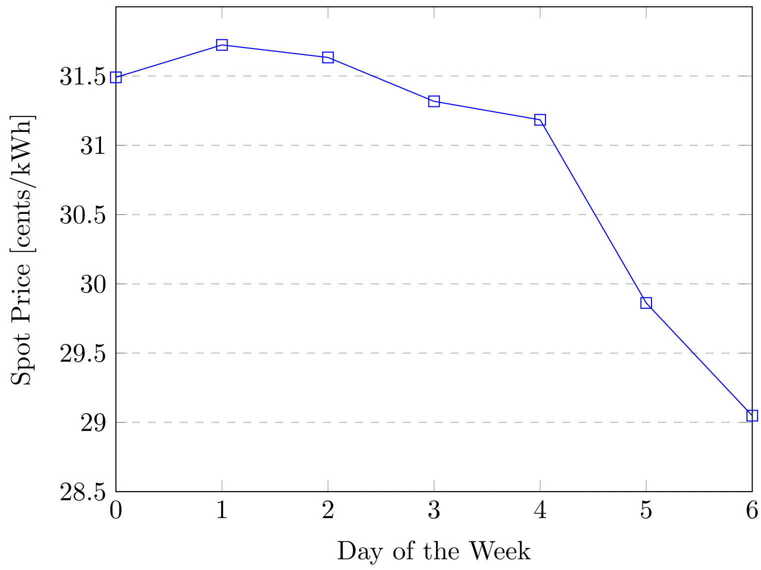

Figure 13: Daily Average Spot Price over the Years

5. Lastly, as it is done with the `day_of_week`, two periodic variables are introduced to represent the day of the year, using sine and cosine functions: `day_of_year_sin` and `day_of_year_cos`. This allows the model to better capture seasonal patterns within the data, illustrated in Figure 12 and 14.

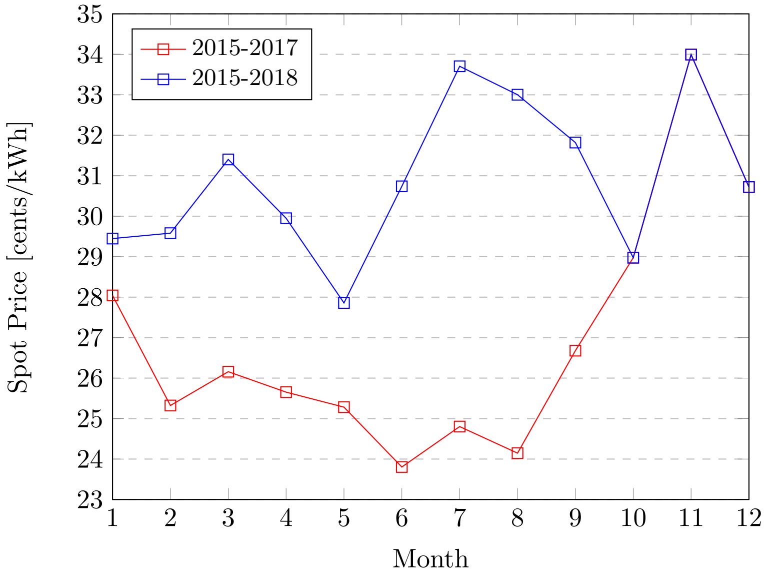

Figure 14: Monthly Average Spot Price/Year Comparison

Furthermore, as shown in the Figure 14, since the dataset is from 31-12-2015 to 13-09-2018, plotting the data for the entire period reveals a higher monthly average of the `spot_price`, due to a general price increase in 2018. In the graph (blue line), this increase occurs only from January (1) to September (9). Because data for the entire year 2018 is unavailable, the last four months (October (10), November (11), and December (12)) show the same monthly average for 2018. The decrease in monthly average spot price is illustrated by the red line, which represents data excluding 2018.

To conclude, in Figure 15, the final correlation matrix is shown:

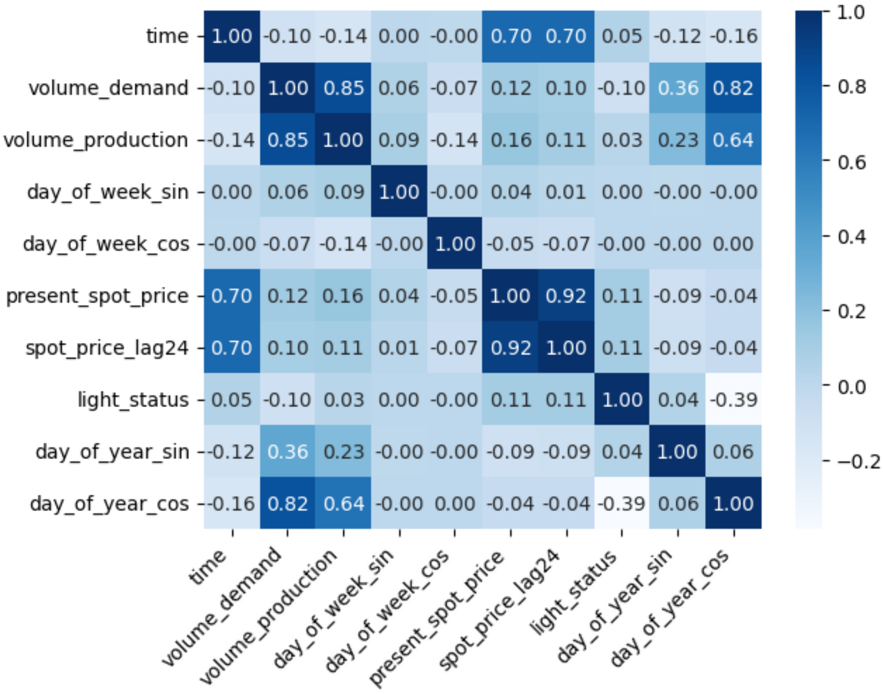

Figure 15: Correlation matrix of the final combination of features

The final correlation matrix after the Feature Engineering shows how the new features have impacted the relationships between variables compared to the initial dataset, which contains only `datetime`, `spot_price`, `volume_demand`, and `volume_production`. A rundown analysis of how these new features have influenced the correlations is listed below:

1. **Sinusoidal transformations for day of the week and day of the year**: these features capture weekly and yearly seasonality, improving the ability to reflect the periodic relationships. For example, `day_of_year_cos` shows a relatively high correlation with `volume_demand` and `volume_production`, suggesting an annual seasonal effect on demand and production;

2. **Lagged spot price**: this variable has a very high correlation with `present_spot_price`, so the model gains a feature that reflects the persistence of spot price trends over time;

3. **Introduction of light status**: the new feature `light_status` provides useful information for daily and night-time patterns.

3.5 Models

Linear Regression

The Linear Regression model is a statistical technique used to estimate the relationship between a dependent variable, or response, (in this case the day-ahead electricity spot price) and one or more independent variables (or predictors). The goal is to find the best-fitting linear equation that predicts the dependent variable based on the independent variables. The relationship is represented by a line, defined by the equation $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \epsilon$, where y is the dependent variable, x_i are the predictors, β_i are the corresponding coefficients and ϵ is the error term [14].

XGBoost

XGBoost is an enhanced decision tree model. The term "*enhanced*" or "*boosting*" refers to a family of algorithms that transform weaknesses in learning into strengths. Boosting is an ensemble learning technique used to reduce bias and variance [15]. In practice, this model works by learning from a series of decision trees, where each tree is trained on a subset of data, and the predictions of each tree are combined to produce the overall prediction. During the training process, XGBoost optimizes an objective function that measures the discrepancy between the model's predictions and the actual output target values. The goal of XGBoost is to minimize this discrepancy to achieve better predictions than using a single decision tree [16].

Random Forest Regression

Random Forest Regression is a robust machine learning method for predicting continuous values that combines the outputs of multiple decision trees to reduce overfitting and improve prediction accuracy. Indeed, Random Forest employs a method known as Bootstrap Aggregation, or bagging, which combines the predictions of multiple decision trees built from randomly selected subsets of training data to produce the final output, rather than relying on just one tree.

3.6 Error Metrics

To evaluate the quality of the forecasts related to the sale price of electricity provided by the implemented machine learning model, different metrics can be used to quantify the error committed. However, in order to better interpret the estimates, it is necessary to use more than one quantity to calculate the error. The most important metrics on which this performance analysis focuses are RMSE, MAPE, and MAE. They provide complementary visions and allow a more in-depth exploration of the predictive capabilities of the built algorithm.

- **Mean Absolute Error (MAE):**

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{1}$$

  The interpretation of the value assumed by the Mean Absolute Error (MAE) can be more or less significant depending on the context. A "small" value of the MAE index can be explained by the fact that all the forecast errors relating to the N instants of time considered in the sum are close to zero in absolute value. On the other hand, the index, as it is defined, does not allow to compare the size of the error with the quantity to be estimated, or in this case the actual price of electricity at a given instant of time. In this sense, there is a certain arbitrariness in the definition of the threshold below which the MAE can be considered reasonably "small" and the interpretation is therefore not completely objective.

- **Mean Absolute Percentage Error (MAPE):**

$$\mathrm{MAPE} = \frac{1}{N}\sum_{i=1}^{N} \left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{2}$$

  In order to circumvent the drawbacks associated with the use of MAE in the application field, another index for quantifying uncertainty can be used, namely MAPE. The Mean Absolute Percentage Error measures the average deviation of the forecasts, expressed as a percentage. This means that the absolute value of the difference between the estimate provided by the model and the actual price observed on the market a posteriori is compared with the actual price. This index also has practical limits due to the fact that overestimates weigh more than underestimates. In other words, overestimation is penalized more than underestimation and this metric turns out to be ineffective when the quantity to be estimated is close to zero (in this case the price of electricity will never be zero and therefore there is no risk of dividing by zero). Since the values are expressed as a percentage, it is possible to normalize them in such a way as to guarantee that they belong to the interval [0,1]. It is the only index among the three proposed that enjoys this property, while the other two indices are not limited and there are no bounds for the values they can assume.

- **Root Mean Square Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \tag{3}$$

The Root Mean Square Error deserves a separate discussion. Unlike other indices, to ensure that the result is a positive value, the squares of the differences are considered and this could lead to the risk of making the sum of these quadratic deviations very "large". For this reason, the square root of the sum is considered so that the result is interpretable and has the same unit of measurement as the quantity to be estimated. The more the forecast differs from the actual price, the greater the penalty brought about by this metric, which will therefore be less sensitive to "small" forecast errors. If the historical series contains anomalous sales data and it is expected that the machine learning model will not be able to pick up on these irregularities, then it is more appropriate to use the MAE as an index of goodness, since the latter is more robust to occasional and non-recurring sales prices and can help for further analysis.

4 Evaluation and Interpretation

This section presents the results obtained using the final forecasting model. In general, as shown in Figure 16, the models are built using a random split of the original dataset (orange color) with the last 24 observations set aside, and with the drop of the first 24 observations used to create the new feature `spot_price_lag24`. This split is performed randomly through Python's `train_test_split` function, allocating 80% of the data for training (blue color) and 20% for testing (green color). Those final 24 rows, removed before the initial split, serve as an extra test set (grey color) to evaluate the models' predictive accuracy specifically for the subsequent 24-hour period, specifically, from 2018-09-11 02:00 to 2018-09-12 02:00 (red color).

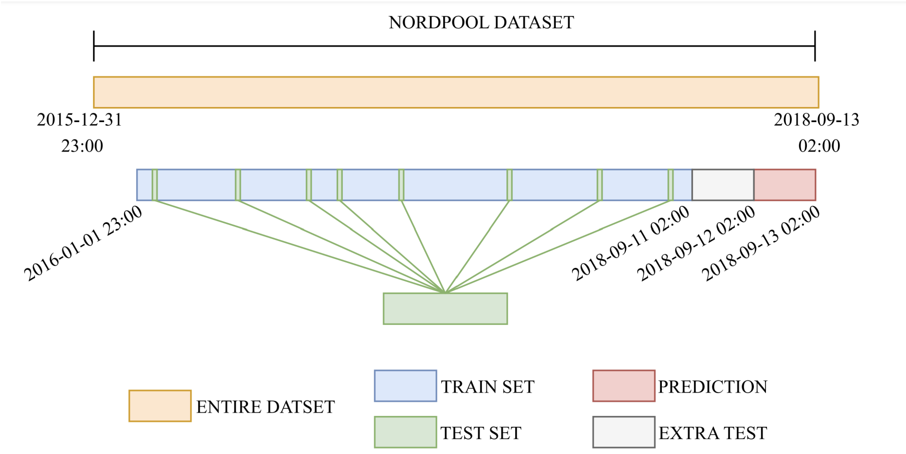

Figure 16: Logic of the Prediction

Moreover, the work is organized into three phases, differentiated by the combinations of features used. Additionally, for each phase, the three models described in Subsection 3.5 are evaluated.

Starting with the **first phase**, only the features originally present in the Nord Pool dataset are selected, namely `time` (in datetime format), `volume_production`, `volume_demand` and `present_spot_price`, to forecast the price for the same time slot on the following day.

Next, in the **second phase**, lagged observations of `present_spot_price` (24 hours) and two periodic features representing the day of the week are introduced, to provide the model with additional information.

Finally, in the **third phase**, a variable representing sunlight and two new periodic variables related to the day of the year are added. The generation of these variables is illustrated in Subsection 3.4.

4.1 Error Analysis

In order to evaluate the models performances, the error metrics described in Subsection 3.6 are computed for the three different stages of the preprocessing procedure.

Model	Linear Regression	XGBoost	Random Forest
MAE	1.98	1.52	1.29
MAPE	7.03	5.70	4.61
RMSE	3.74	2.84	2.66

Table 4: Errors - First phase

Model	Linear Regression	XGBoost	Random Forest
MAE	1.97	1.26	1.16
MAPE	7.05	4.48	4.13
RMSE	3.67	2.23	2.39

Table 5: Errors - Second phase

Model	Linear Regression	XGBoost	Random Forest
MAE	1.97	1.13	1.07
MAPE	7.08	4.02	3.81
RMSE	3.67	2.04	2.23

Table 6: Errors - Third phase

Both a comparison between the three phases and within each stage can be done. The first kind of analysis highlights that the last combination of features produces the best result in terms of the three metrics for both XGBoost and Random Forest, whereas Linear Regression achieves similar values of the metrics in all the three stages. In general, the tree-based models clearly benefit from the introduction of new features, resulting in lower values of the metrics.

Focusing on each table, it is clear that Random Forest outperforms the other two models in all the metrics, apart from the RMSE of the last phase, that is slightly bigger than the one achieved by XGBoost.

4.2 Feature Importance

In the following subsection, given the error analysis, the feature importance of the two primary models developed - XGBoost and Random Forest - are presented.

1. **Random Forest regression** (Figure 17): the model's predictions are primarily driven by the feature `present_spot_price`, which has an importance score of 0.8625, indicating it holds the most significant predictive power and closely correlates with the target outcome. While time follows as the second most influential feature with a much lower importance score of 0.0367, it still provides notable information. The `spot_price_lag24` feature, representing the lagged spot price from the previous day, has an importance score of 0.0232, hinting at a correlation between consecutive days. Other variables, such as `volume_demand` (0.0191), `day_of_year_cos` (0.0145), `volume_production` (0.0146), and `day_of_year_sin` (0.0119), make smaller contributions, reflecting lower but still relevant influences. Finally, `day_of_week_cos` (0.0089), `day_of_week_sin` (0.0047), and `light_status` (0.0040) are the least impactful features, suggesting they contribute minimally to the predictive power of the model. This distribution of importance highlights the fundamental role of `present_spot_price`, while the other features provide supplementary, though progressively weaker, predictive value.

2. **XGBoost regression** (Figure 18): the comparison of XGBoost feature importance with the previous model's ones highlights some significant dynamics. The feature `present_spot_price`

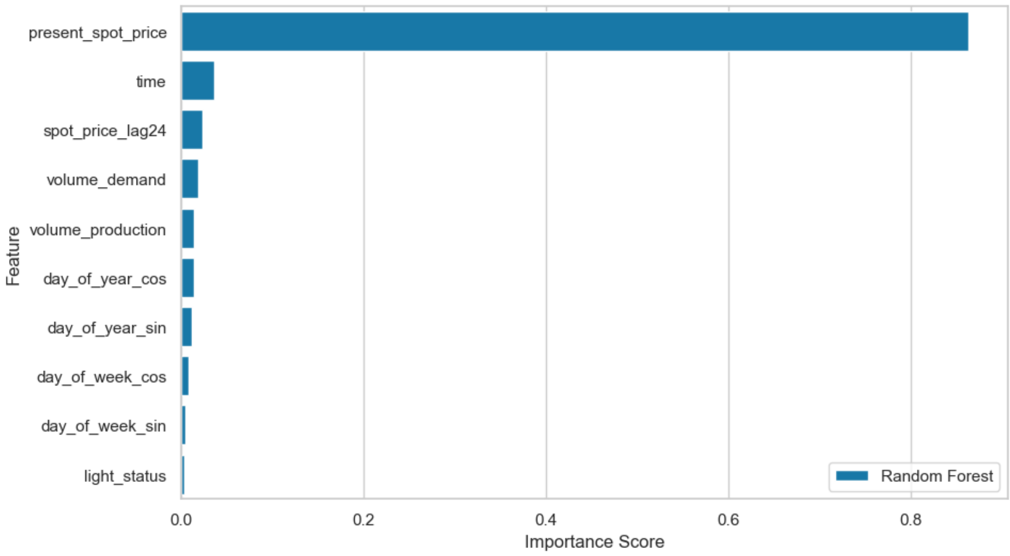

Figure 17: Random Forest Feature Importance

remains the most influential, although its importance has slightly decreased from 0.8625 to 0.8056. This suggests that while it continues to be crucial for predictions, there may be greater diversification or an evolution of relationships between features in the new model. Other features, such as `day_of_week_cos` and `day_of_year_cos`, show an increase in their importance compared to the previous model, indicating that temporal information has gained a more relevant role. Surprisingly, the feature `time` has seen a decrease in its importance, from 0.0367 to 0.0294, suggesting it may have a lesser impact in the new configuration of the model. The decrease in importance of features like `volume_demand` and `volume_production` is particularly important, as it reflects a non-standard relationship between supply and demand in relation to prices.

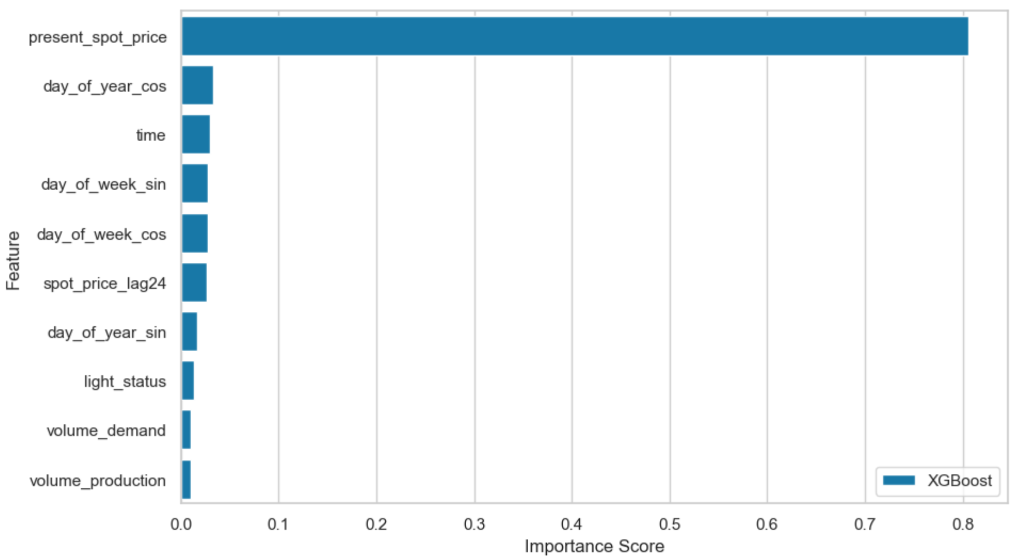

Figure 18: XGBoost Feature Importance

4.3 Compare the three models

Analyzing the performance of the three models - Linear Regression, XGBoost, and Random Forest - it is clear how the prediction accuracy changes across the three stages of feature combinations for each model analyzed. As shown in Tables 4, 5, and 6, Random Forest outperforms Linear

Regression and XGBoost on all metrics (MAE, MAPE, and RMSE) at each stage, except from the RMSE at the final stage, where XGBoost gets a better result. The tree-based models, especially Random Forest, show an improvement in performance when new features are introduced, meaning that they benefit from the additional temporal and periodic information. Meanwhile, Linear Regression's metrics remain unchanged indicating that a limited impact from feature engineering.

To see these differences, Figure 19 presents the MAE, MAPE, and RMSE values for each model in the last phase.

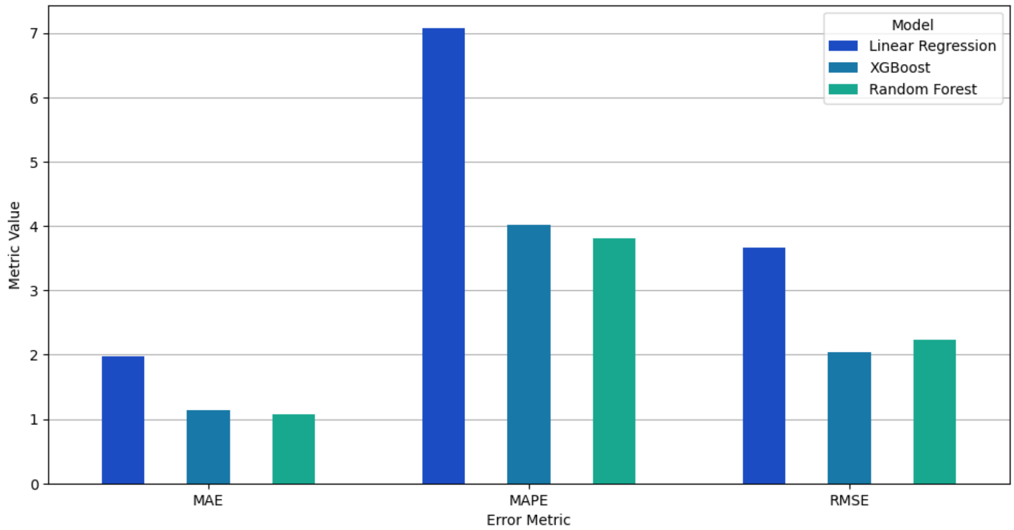

Figure 19: Compare Errors

Additionally, the feature importance of the two primary models developed for the final stage shows that both XGBoost and Random Forest assign the highest weight to the `present_spot_price`, confirming its predictive value in forecasting electricity spot prices. However, XGBoost has a more diverse distribution of feature importance, with time-based features like `day_of_week_cos` and `day_of_year_cos`, gaining increased importance. This suggests that XGBoost captures additional temporal dynamics compared to Random Forest, although it is still worse in overall performance.

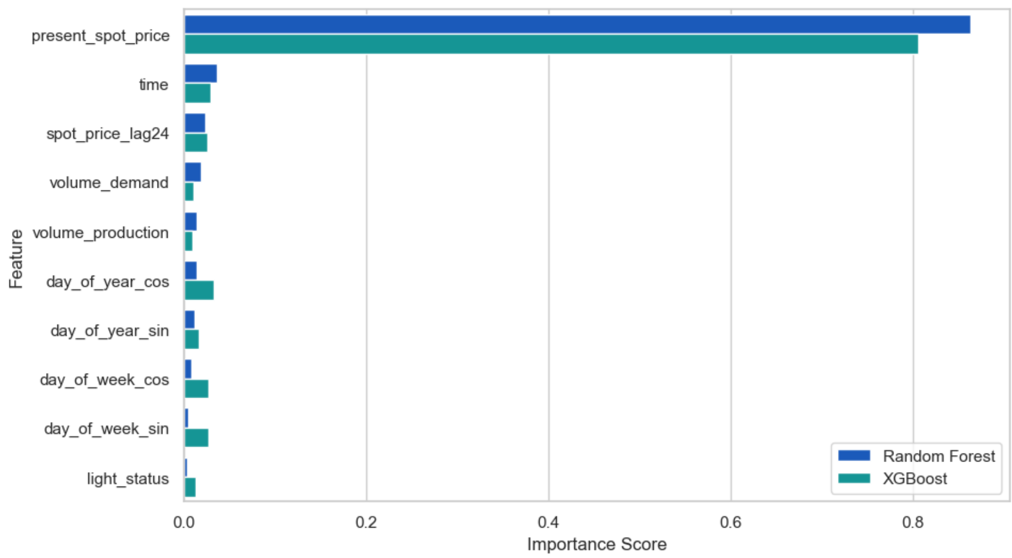

Figure 20: Compare Feature Importance

In summary, Random Forest is the most accurate model for predicting Nord Pool electricity spot prices, followed closely by XGBoost. Linear Regression is more straightforward and easily interpretable, but lacks in the predictive power needed for effective forecasting in this context. In particular, Random Forest is a valuable tool for stakeholders seeking to anticipate price trends in highly dynamic energy markets.

4.4 Results

Random Forest is chosen as the final model for prediction in the context of this project. The results of the predictions are shown in the Figure 21.

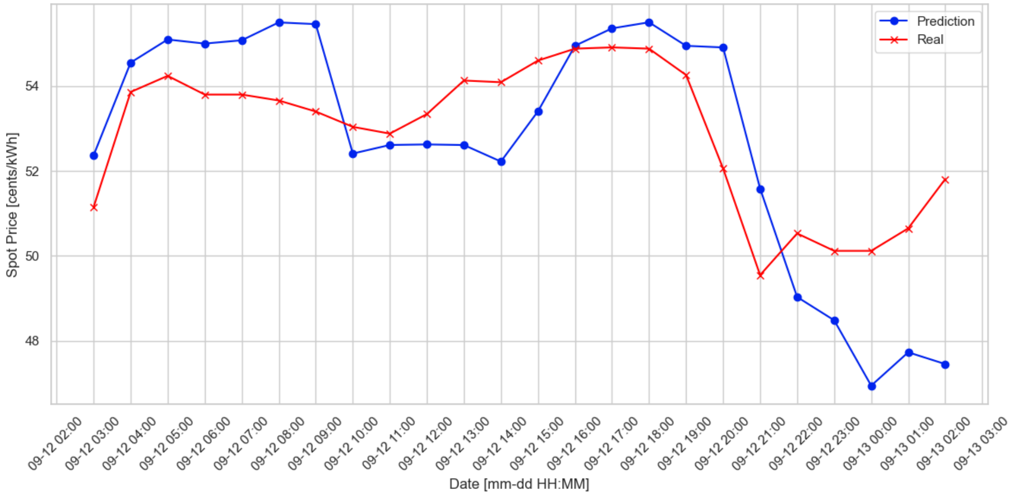

Figure 21: Predictions vs Real Values

It is possible to notice that the predictions have a similar trend with respect to the real data. In addition, the forecast values have an higher upper bound and a smaller lower bound with respect to the true values, probably due to the fact that the training data do not contain a well defined pattern through the years, as highlighted in Figure 14.

4.5 Business Value

The potential benefits of an effective predictive model for an energy trader are countless. In this section, they are summarized in three points:

1. **Optimization of bid/sell strategies**: being able to anticipate eventual price drops or peaks may provide the trading agency with fundamental information for the development of a very profitable strategic plan. Trivially, this translates in buying energy when prices are expected to decrease, while selling where prices are expected to drop.

   This kind of model can also be integrated with an automatic trading algorithm, which might allow operations on a large-scale, reducing human impact, while improving efficiency and fastness;

2. **Risk mitigation and losses prevention**: thanks to a predictive model, traders may be able to elaborate better hedging strategies, possibly preventing big losses generated from unexpected sudden price changes. This leads to a big competitive advantage with respect to others market actors who can not rely on effective predictions;

3. **Partnerships with industrial costumers and large companies**: traders can establish agreements with companies that have flexible loads, such as factories, to coordinate consumption based on price forecasts. An accurate model allows for rigorous planning of these load reductions or increases, improving efficiency and generating wealth for both parties through the agreement.

5 Deployment and Recommendations

This section serves as a baseline for actualizing the financial aspect of the project, as it is of vital importance to be able to sell the product just drafted, outlining potential buyers and parties of interest and creating a framework for a deployment plan and different strategies to do that.

5.1 Deployment Plan

The process begins with establishing a robust **data pipeline**, in which three main tasks are outlined: **data collection**, **preprocessing**, and finally **storage**. To do specifically that, an ETL process (Extract, Transform, Load) is set up, where raw data are taken in from the market, then normalized and transformed in a useful form for businesses, and finally stored securely in a database, while taking into consideration storage capacity and quick data retrieval too.

The next step is **model training and testing**. Using the data made available by Nord Pool, extensive training and testing are done to aim for best performance. Note that the model shall be capable of being re-trained when new data becomes available, to provide continuous improvement and better accuracy over time.

To allow clients to use the predictions made by the model, a tailor-made **API** shall be developed to allow access through a secure web connection. The model should be hosted on a cloud platform such as Google Cloud or AWS to ensure fast responses and the ability to scale up when the number of clients grows, and containers shall be used to easily deploy different model updates while storing older versions.

A user-friendly **dashboard** shall be developed to ensure positive client interaction, providing visual feedback on the predictions, historical analysis, and confidence intervals. It is important to make the dashboard customizable and provide clients with the ability to export the data.

Special attention must be given to **security**, as data should be encrypted, and an access protocol should be implemented to ensure the predictions are theft-proof. It is necessary to pay attention to privacy regulations too, as outlined previously.

Client support have to be developed: to make the product usable, it is necessary to produce a user guide, tutorials for using the product, and short training sessions provided by a support team to assist with navigating the platform. It is also important to provide live real-time assistance in case issues arise and prevent clients from using the product.

In the end, it is important to implement a **feedback loop** to acquire client feedback, which will guide improvements and changes. System logs and performance must be monitored to detect any issues for quick fixes, all to ensure the product's long-term usage.

5.2 Relevant Actions Towards Different Stakeholders

In this project, the primary stakeholders are identified as **Retail Energy Suppliers and Traders**, who rely on timely and accurate energy price forecasting to make effective pricing and purchasing decisions. The developed Random Forest model supports these stakeholders by offering demand projections, facilitating rapid adjustments to market changes, and enabling custom alerts for significant price movements. These features are essential for competitive positioning in a highly dynamic market. Retail energy suppliers benefit from the system's API integration, which allows them to automate pricing adjustments according to forecast price fluctuations. Traders, on the other hand, utilize the model's demand projections and real-time data access to optimize trading decisions, ensuring they are well-prepared to respond to sudden price shifts. Custom alerts play a critical role in enabling traders to stay updated on market conditions, giving them the flexibility to act swiftly and maximize profitability.

While retail energy suppliers and traders are the main stakeholders in the current implementation, this model can be expanded to meet the needs of other potential stakeholders with different requirements:

- **Utility companies and energy providers:** these type of clients are looking for robust forecasting models, so there will be the need to integrate already pre existing energy management systems, and frequent data updates shall be constantly provided for scalability. The

API and dashboard customization come into play to access granular data, forecasts that are specific to some regions and historical analyses;

- **Large industrial and manufacturing corporations:** A simplified and intuitive interface shall be the focus, since the main objective of these clients is to minimize costs, not analyzing market mechanics. Alerts for peak and off-peak pricing could be added to catch the attention more easily on only the relevant information. Periodic updates shall be offered instead of fast ones, since industrial clients need to mainly make high level decisions. It is also important to highlight low cost energy windows for the purpose of letting them align their schedule with them;

- **Energy traders and hedge funds:** they require immediate access to the latest data and predictions, so it is important to prioritize fast and live data delivery and put extra focus on quick response time. It is also important to introduce more analysis tools, like trading indicators, volatility projections and risk analysis tools for them to be able to look at. It is useful to provide the option for the client to back test predictions with past data, and perform simulations of said historical data to test hypothesis and develop new strategies;

- **Grid operators and energy management companies:** It would be useful to offer integration with systems used for load balancing and grid flow management, with the intention of providing in border localized predictions, and cross border price forecasts if they manage multiple regions or countries. Implementing alerts for spike in supply and demand are is also something to take into consideration, to allow them to respond quickly to grid instabilities;

- **Governmental and regulatory bodies:** emphasis shall be put on long-term forecasting to support policy planning and market stability efforts, as their focus is much more on planning rather than quick-time decision making. Extra focus shall be put on making sure that models meet regulatory transparency requirements by providing detailed model explanations and variable tracking, while also enabling exportable and anonymized reports to be shared with government departments and the public at the same time;

- **Green energy producers and initiatives:** it is important to include renewable energy metrics, such as solar or wind forecasts, to provide insight on when to store energy versus feeding it into the grid based on price predictions with the scope of maximizing revenue. Generating reports or graphic visualizations for grid stability and emission shall also be of focus, to align with environmental reporting.

In summary, while retail energy suppliers and traders form the core focus of this project, the versatility of the forecasting model makes it well-suited for expansion to serve a diverse array of stakeholders in the energy market.

5.3 Limitations and Improvements

Dataset Limitations and Improvements

One of the most delicate phases in the field of electricity price forecasting is feature engineering. Most of the work must therefore focus on data preparation since the raw dataset initially provided may not be suitable for the implementation and training of a Machine Learning model, especially if working with historical series. To have more accurate estimates of the future exchange price on the electricity market (sale/purchase) it is essential to adopt techniques for manipulating the main characteristics of a historical series such as trends, seasonality, variations in variability, outliers and/or ensuring that the time scale/sampling instants are adequate. If necessary, you can consider performing a data normalization to ensure that the values of some variables belong to a specific appropriate range of values.

The dataset of Nord Pool is characterized by 4 columns and the variables available may not be sufficient to achieve the long-term strategic objectives of the stakeholders. Furthermore, the dataset has a relatively "small" number of rows compared to typical datasets used to train automatic learning algorithms (23667 rows) and this could complicate the operation of dividing the dataset between training set and test set. The usual cross-validation techniques could therefore fail due to the scarcity of data available and it is necessary to resort to computationally more complex methods, for example by dividing the dataset into blocks and performing the cross-validation several times with the only constraint relating to the fact that the training data must chronologically precede the testing data.

Before providing the dataset to the built Machine Learning model, it may be necessary to integrate the original data by adding new variables considered relevant in order to make accurate predictions. In other words, to solve the problem related to the few variables present in the original dataset, it is necessary to rely on safe, reliable and proven professional centers that deal with the collection of values of quantities related to electricity consumption such as, for example, temperature and hours of light/darkness. This is what it is done during Feature Engineering.

Method Limitations and Improvements

Once the dataset has been cleaned of any imperfections, most of the work is done. The next step is choosing the model, which can often be complicated since there are different algorithms and they can provide very different estimates from each other. To make an informed choice, two complementary filtering criteria can be applied: first, all those models whose predictions are wrong can be discarded, that is, those that make an error above a certain threshold considered acceptable, and second, the "least expensive" model is chosen among those remaining.

To compare the performance of different algorithms, there are different libraries for different programming languages. One of these is PyCaret, available in Python, now the programming language par excellence in the Machine Learning field. Libraries of this type perform the work automatically after setting some characteristic parameters and providing the dataset in a standardized format. The advantages are several: with a few simple lines of code it is possible to test the predictive capabilities of multiple models simultaneously based on the previously defined Error Metrics and the programming part is therefore reduced to a minimum, it is possible to achieve better results by combining the qualities of multiple algorithms and compensating for defects. In this context it is essential to define what is meant by a better model. From a theoretical point of view, a model is considered better if, with the same accuracy in the prediction, it uses a smaller number of resources or uses the same availability but for a shorter period of time.

From this it can be deduced that, at the company level, in addition to determining a reasonable acceptance threshold for the model estimates, a compromise must also be found between the accuracy of the forecast itself and the work in the broad sense used to obtain such estimates (including manpower hours and the economic cost of the resources/energy used to allow the model to function in the best possible way. Among the cost items to be considered are those related to the use of information systems and scalability. Since electricity prices can also depend on geographical characteristics specific to the area under consideration, it is useful to have different models available, each of which deals with a region/district of interest. Distributed systems for the control of different models help to achieve this desired company structure at a minimum cost for the services offered. The companies that provide these services offer customized packages depending on the needs and, if necessary, allow the plan to be upgraded to obtain greater potential.

Another key improvement is hyperparameter tuning, which involves optimizing the settings (or "hyperparameters") that control the behavior of a machine learning model. Unlike model parameters, which are learnt from the data during training, hyperparameters are set prior to training and can significantly impact the model's performance. By systematically testing and adjusting these hyperparameters, it is possible to enhance the model's predictive accuracy, reduce overfitting, and improve its efficiency.

5.4 Future Analysis

Future analysis must be taken into consideration to complete what has been described in the previous sections of this report and to understand how the results obtained can be validated over long-term time horizons. For the organization of possible interventions, it is appropriate to deepen the following topics of interest:

- **Search for the best analysis configuration**: training multiple models simultaneously using automated tools speeds up the decision-making process for implementing the application that supports the electricity market. The optimization of performance comparison operations also allows for quantitatively understanding the impact of each modification made and evaluating the effectiveness of the best model;

- **Learning from new, unseen data**: the types of variables added each time must always be the same (their meaning should not change over time); otherwise, the implemented model may, at best, fail to work or even provide inconsistent estimates because it gets "confused" by the presence of variables different from those it was trained on;
- **Integrated real-time data collection systems**: as mentioned earlier, in order to ensure that the model does not learn any biases, in addition to capturing the fundamental aspects for analyzing a time series, it is essential to integrate the available information with additional data from reliable sources. In the absence of business partners responsible for collecting these additional variables, specific IoT tools (such as for measuring external temperature, for example) can be used. This way, it is also possible to account for unforeseen events that would otherwise only be detectable after more or less variable time windows;
- **Collecting feedback from stakeholders**: the feedback from users who will use the machine learning application to extract useful suggestions on how to behave in the market and understand the evolving dynamics of electricity prices is crucial, as the objectives pursued may change over time and needs are not constant. Their input can suggest a change of direction and provide valuable guidance on how to "alter" the original model to create updated versions of the application and meet new requirements.

6 Monitoring and Maintenance

Monitoring and maintenance are essential to ensure the ongoing reliability and performance of the data analysis system. This section lists the strategies and tools used to track system health, evaluate performance, and address potential issues. Regular monitoring helps detect anomalies, assess the quality of predictions, and optimize the system based on real-time feedback.

The subsections that follow provide an overview of key components for operational success. In **Key Performance Indicators**, metrics that measure the effectiveness of the developed system and the quality of its outputs are identified. The **Monitoring Dashboard** describes the user interface and tools used to visualize and track these KPIs in real-time. Finally, the **Strategic Risk Management and Contingency Plan** explains the approach for identifying, assessing, and mitigating risks, ensuring the system remains resilient and adaptable in case of unexpected challenges.

6.1 Key Performance Indicators

Model Performance KPIs

The following Key Performance Indicators (KPIs) are defined to assess the performance of the machine learning model developed for forecasting electricity prices in the Nord Pool market, in fact these KPIs are selected to ensure that the model developed not only provides accurate predictions but also meets standards of robustness, efficiency, and fairness. Monitoring these KPIs over time enables continuous improvement, allowing identification of areas where the model may require further tuning or adjustments.

- **MAE**
  - *Description*: Measures the average size of the mistakes in a collection of predictions, without taking their direction into account.
  - *Rationale*: Low MAE indicates effective model performance and helps build trust in the predictions generated.
  - *Metric*: It is measured as the average absolute difference between the predicted values and the actual values.
  - *Threshold*: MAE score below 1.5.

- **MAPE**
  - *Description*: Computes the average deviation of the forecasts, expressed as a percentage.
  - *Rationale*: MAPE gives a better idea of the relative error committed than MAE, since each absolute difference is divided by the true value of the observation.
  - *Metric*: It is computed as the average of the mean absolute differences, each one divided by the actual value.
  - *Threshold*: MAPE score below 4.
- **RMSE**
  - *Description*: Provides a balanced measure of prediction accuracy, with sensitivity to larger errors.
  - *Rationale*: RMSE is useful when large errors need stronger penalties.
  - *Metric*: It is computed as the square root of the mean squared differences between predicted and actual values.
  - *Threshold*: RMSE score below 2.5.
- **Model Robustness**
  - *Description*: Assesses the model's ability to perform under varied or noisy input conditions.
  - *Rationale*: Robust models maintain reliability across a range of conditions, ensuring dependable performance.
  - *Metric*: Performance degradation under perturbed inputs.
  - *Threshold*: Less than 10% performance drop under controlled noise or input perturbations.
- **Processing Speed and Latency**
  - *Description*: Measures the time taken for the model to process an input and return a prediction.
  - *Rationale*: Essential for real-time applications where fast response is required.
  - *Metric*: Average response time per input in milliseconds.
  - *Threshold*: Target response time below 500 ms for real-time applications.
- **Scalability**
  - *Description*: Evaluates the model's ability to handle big data volumes without significant performance loss.
  - *Rationale*: Scalability is crucial for models expected to operate in environments with fluctuating data loads.
  - *Metric*: Performance stability across varying data loads.
  - *Threshold*: Consistent performance under up to a 10X increase in data volume.

Each KPI contributes to an evaluation of the model, ensuring it meets performance, robustness, and operational standards necessary for reliable electricity price forecasting in the Nord Pool market.

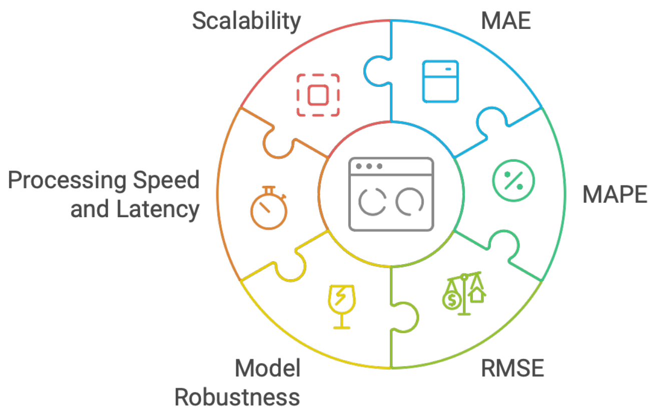

Figure 22: Model Performance Evaluation

Business KPIs

In addition to evaluating model performance, it is crucial to evaluate the business impact of the forecasting model in the Nord Pool market. The following Business KPIs are defined to measure the model's contributions to strategic business objectives, financial outcomes, and decision-making effectiveness, in fact these KPIs help ensure that the model's outputs are not only technically correct but also valuable from a business perspective.

- **Forecasting Accuracy Impact**
  - *Description*: Measures the degree to which accurate forecasts improve operational and trading decisions.
  - *Rationale*: Higher forecasting accuracy enables better-informed decisions and strategies, potentially increasing the profit.
  - *Metric*: Improvement in decision outcomes or profitability due to forecast accuracy, e.g., percentage increase in successful bids or reduction in financial losses.
  - *Threshold*: Target a minimum 5% improvement in outcomes due to more accurate forecasting.
- **Revenue from Optimized Bidding Strategies**
  - *Description*: Evaluates the impact of forecast-driven bidding strategies on revenue generated in the Nord Pool market.
  - *Rationale*: Accurate price forecasts allow for more competitive and effective bidding strategies, directly influencing revenue.
  - *Metric*: Percentage increase in revenue attributable to forecast-based bidding improvements.
  - *Threshold*: Target a 7% increase in revenue through optimized bidding practices enabled by the model.
- **Risk Mitigation Effectiveness**
  - *Description*: Measures the model's effectiveness in supporting risk management strategies, particularly in mitigating exposure to price volatility.
  - *Rationale*: Better risk management improves financial stability by minimizing potential losses in volatile market conditions.
  - *Metric*: Reduction in financial exposure due to improved risk management, e.g., decrease in unexpected cost fluctuations.
  - *Threshold*: Achieve a 10% reduction in financial risk exposure with the use of accurate forecasting.

- **Stakeholder Satisfaction**
  - *Description*: Measure the satisfaction levels of the main stakeholders, such as retail suppliers and traders, with the model's impact on decision-making and market transparency.
  - *Rationale*: High satisfaction indicates that the model meets business needs and enhances stakeholder confidence.
  - *Metric*: Satisfaction score from stakeholder feedback or surveys.
  - *Threshold*: Target a minimum satisfaction rating of 85% from key stakeholders.

These Business KPIs ensure that the model's contributions align with financial, operational, and strategic goals. By focusing on cost savings, revenue growth, risk mitigation, and stakeholder satisfaction, these KPIs help measure the real-world value generated by the forecasting model developed in the Nord Pool market.

6.2 Monitoring Dashboard

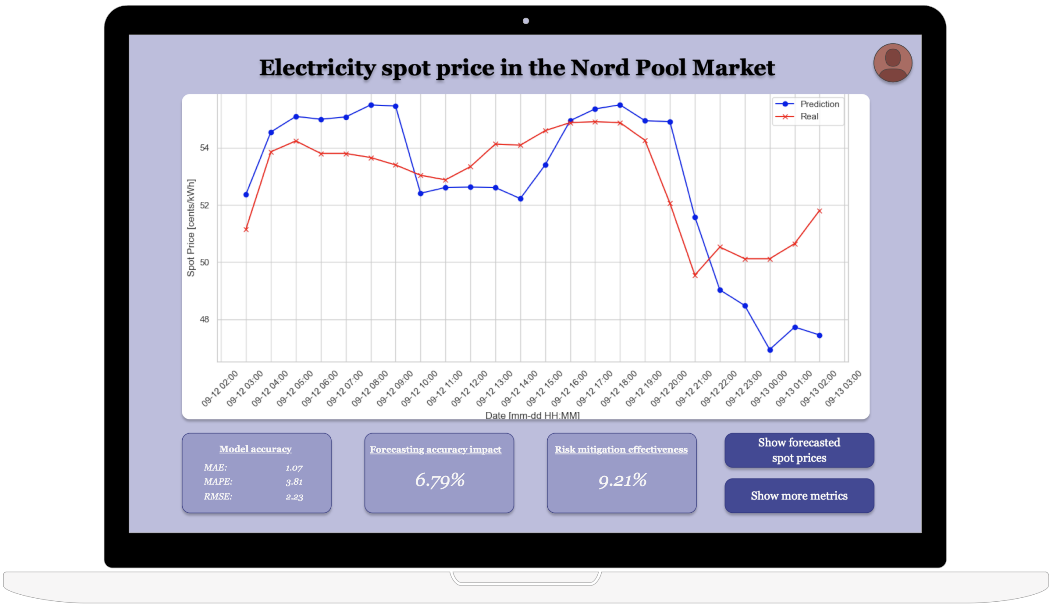

Figure 23: Monitoring dashboard

The monitoring dashboard provides a visual overview (GUI - graphic user interface) of the model's performance and key metrics for tracking electricity spot prices in the Nord Pool market. On the top, the dashboard displays a line chart comparing the actual spot prices (in red) and the model's predictions (in blue), offering a clear view of the model's forecasting accuracy over time. Below the chart, a series of key performance indicators (KPIs) lists critical metrics, including metrics such as MAE, MAPE and RMSE, forecasting accuracy impact, and risk mitigation effectiveness.

These metrics provide valuable insights into the model's reliability and its ability to manage potential risks in the electricity market, additionally, interactive buttons on the right allow users to explore further details, such as forecasted spot prices and viewing additional metrics. Overall, this dashboard serves as a tool for monitoring model performance and assessing its alignment with business objectives in real-time.

6.3 Strategic Risk Management and Contingency Plan

Managing risks effectively is crucial to ensuring the reliability and resilience of the forecasting model in the Nord Pool market. This section outlines key strategic risks associated with the model deployment and operation, as well as contingency plans to mitigate those risks.

- **Data Quality and Integrity Risks**
  - *Risk*: Inaccurate or incomplete data can lead to unreliable forecasts, impacting decision-making and strategic actions for stakeholders.
  - *Mitigation*: Implement real-time data validation processes to detect and correct data anomalies, with regular verifications and quality checks of the data sources ensure high-quality inputs.
  - *Contingency Plan*: Establish a secondary data source to replace or complement the primary source in case of data quality issues.
- **Model Drift and Performance Degradation**
  - *Risk*: Over time, the model may become less accurate due to changes in market conditions, demand-supply dynamics, or external factors.
  - *Mitigation*: Monitor model performance through established KPIs and implement periodic model retraining with new and updated data to ensure ongoing accuracy.
  - *Contingency Plan*: Deploy an alert system that flags significant drops in model performance and allows for immediate model retraining or replacement with alternative models.
- **Scalability and System Load**
  - *Risk*: The model may face challenges handling large-scale data or a high number of simultaneous requests, especially during peak times.
  - *Mitigation*: Use cloud-based infrastructure with auto-scaling capabilities to adapt to varying loads, ensuring system responsiveness and availability.
  - *Contingency Plan*: Prepare a load-balancing mechanism and additional server capacity to quickly manage unexpected increases in demand.
- **Cybersecurity and Data Privacy Risks**
  - *Risk*: Unauthorized access to the forecasting system or data breaches could compromise sensitive market information and user trust.
  - *Mitigation*: Implement robust encryption, multi-factor authentication, and access control policies to secure data and model access.
  - *Contingency Plan*: Develop a response plan that includes incident detection, containment, and reporting procedures, along with regular system security assessments and updates.
- **Regulatory and Compliance Risks**
  - *Risk*: Changes in market regulations or data privacy laws may require adaptations to the model or its data handling processes.
  - *Mitigation*: Maintain active engagement with regulatory updates and ensure model documentation and processes are compliant with current standards.
  - *Contingency Plan*: Establish a protocol for rapid adaptation of the model and data processes in response to regulatory changes, minimizing compliance risks.

Through these risk management strategies and contingency plans, the project aims to maintain the model's robustness, ensuring its continued accuracy and reliability in the dynamic Nord Pool market environment.

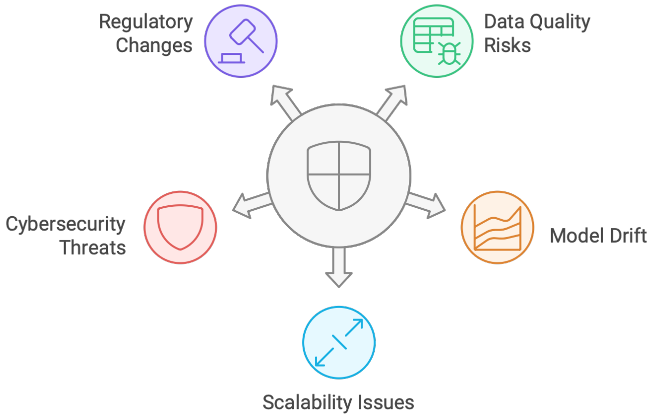

Figure 24: Risk Management in Forecasting Model

6.4 Lessons Learnt and Feedback

The analysis of the Nord Pool dataset and writing this report has given us a lot of insights, both technical and professional, as this was our first experience creating a data science report, in fact we learnt the importance of following a structured approach for a data analysis project, starting from the main analysis to pre-processing and modeling. Understanding each stage's importance helped us to stay organized and make informed decisions throughout the report.

Documenting our work and looking at the results has proven essential for communicating results, this approach also improved our understanding of how to present technical content in a way that is comprehensible with stakeholders who may not have a technical background as we have. In addition, cleaning and transforming the dataset was really challenging, especially understanding which features would contribute with meaning to the model but, through trial and error, we learned how impactful well-crafted features can be on model performance.

Selecting appropriate Key Performance Indicators (KPIs) for model evaluation and identifying business impact was another key learning. This experience highlighted the importance to align technical performance metrics with business objectives to express the model's value. Finally, creating visualizations for data analysis and results points the power of clear graphics in communicating trends, patterns, and outliers. Visualization helped us understanding complex relationships within the dataset and allowed us to point these results effectively.

Overall, this project has provided a solid foundation in data science reporting, providing us with skills and strategies for future analysis. Each stage of the process — data preparation, modeling, evaluation, and communication — offered us new learning opportunities that can guide us improving our ability in next data science projects.

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